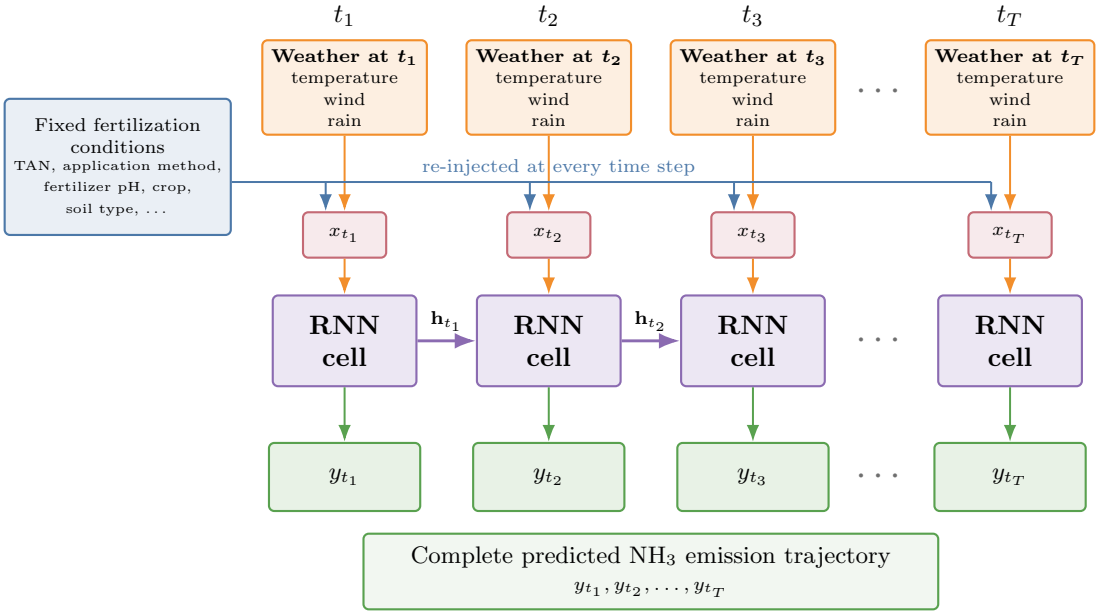

t_1
t_2
t_3
t_T
Weather at t_1
temperature
wind
rain
Weather at t_2
temperature
wind
rain
Weather at t_3
temperature
wind
rain
Weather at t_T
temperature
wind
rain
Fixed fertilization conditions
TAN, application method, fertilizer pH, crop, soil type, ...
re-injected at every time step
x_{t_1}
x_{t_2}
x_{t_3}
x_{t_T}
RNN cell
RNN cell
RNN cell
RNN cell
$\mathbf{h}_{t_1}$
$\mathbf{h}_{t_2}$
y_{t_1}
y_{t_2}
y_{t_3}
y_{t_T}
Complete predicted NH_3 emission trajectory
$y_{t_1}, y_{t_2}, \ldots, y_{t_T}$